

7.9 Classwork: Exponential and Logarithmic Word Problems — Markscheme

1. [Maximum mark: 13]

$$N(t) = 200 e^{kt}, \text{ with } N(2) = 800.$$

(a) $N(0) = 200 e^0 = 200.$ A1

(b) Substitute $t = 2$ and $N = 800$: $800 = 200 e^{2k}.$ M1

$$e^{2k} = 4. \quad \text{A1}$$

$$2k = \ln 4 \Rightarrow k = \frac{1}{2} \ln 4. \quad \text{A1 AG}$$

Note: $k = \frac{1}{2} \ln 4 = \ln 2$ (equivalent forms).

(c) $N(5) = 200 e^{5k} = 200 \cdot 4^{5/2}.$ M1

$$4^{5/2} = 32. \quad \text{A1}$$

$$N(5) = 200 \times 32 = 6400. \quad \text{A1}$$

(d) Set $200 e^{kt} = 5000$, so $e^{kt} = 25.$ M1

$$kt = \ln 25 \Rightarrow t = \frac{\ln 25}{k} = \frac{2 \ln 25}{\ln 4}. \quad \text{M1}$$

$$t = 4.6439 \dots \approx 4.64 \text{ hours (3 s.f.)}. \quad \text{A1}$$

(e) Doubling: $2N_0 = N_0 e^{kt} \Rightarrow e^{kt} = 2.$ M1

$$t = \frac{\ln 2}{k} = \frac{\ln 2}{(1/2) \ln 4} = \frac{2 \ln 2}{2 \ln 2} = 1. \quad \text{M1}$$

$$t = 1.00 \text{ hour}. \quad \text{A1}$$

2. [Maximum mark: 12]

\$3000 at 4.5% per annum, compounded annually.

(a) Value after 1 year = $3000 \times 1.045 = \$3135$. **AG**

(b) $V(6) = 3000 (1.045)^6$. **M1**

$(1.045)^6 = 1.30226 \dots$ **A1**

$V(6) = 3000 \times 1.30226 \dots = 3906.78 \dots \approx \3907 . **A1**

(c) $3000 (1.045)^n > 5000 \Rightarrow (1.045)^n > \frac{5}{3}$. **M1**

$n > \frac{\ln(5/3)}{\ln 1.045} = 11.604 \dots$ **M1**

Smallest integer $n = 12$ years. **A1**

Verification: $V(11) \approx \$4868.55 < 5000$; $V(12) \approx \$5087.66 > 5000$. Award A1 for $n = 12$ regardless of method.

(d) Continuous: $A(6) = 3000 e^{0.27}$. **M1**

$e^{0.27} = 1.30996 \dots$, so $A(6) = 3929.8838 \dots \approx \3929.88 . **A1**

Difference: $\$3929.88 - \$3906.78 = \$23.10$. **M1 A1**

Accept \$23.10 or \$23.11 depending on intermediate rounding.

A1 (presentation: both values stated to nearest cent)

3. [Maximum mark: 7]

$$M(t) = 64 (1/2)^{t/30}.$$

$$\textbf{(a)} \quad M(50) = 64 (0.5)^{50/30} = 64 (0.5)^{5/3}. \quad \textbf{M1}$$

$$= 64 \times 0.31498 \dots = 20.158 \dots \approx 20.2 \text{ g (3 s.f.)}. \quad \textbf{A1}$$

$$\textbf{(b)} \quad 64 (0.5)^{t/30} = 8 \Rightarrow (0.5)^{t/30} = \frac{1}{8} = (0.5)^3. \quad \textbf{M1}$$

$$\frac{t}{30} = 3 \Rightarrow t = 90 \text{ years}. \quad \textbf{A1}$$

$$\textbf{(c)} \quad 5\% \text{ remaining means } M(t) = 0.05 \times 64 = 3.2:$$

$$64 (0.5)^{t/30} = 3.2 \Rightarrow (0.5)^{t/30} = 0.05. \quad \textbf{M1}$$

$$\text{Take ln: } \frac{t}{30} = \frac{\ln 0.05}{\ln 0.5} = 4.3219 \dots, \text{ so } t = 129.66 \dots \quad \textbf{M1}$$

$$t \approx 130 \text{ years (nearest year)}. \quad \textbf{A1}$$

4. [Maximum mark: 11]

$$T(t) = 22 + 64e^{-kt}, \text{ with } T(4) = 54.$$

$$\text{(a)} \quad T(0) = 22 + 64 = 86^\circ\text{C}. \quad \text{A1}$$

$$\text{(b)} \quad \text{Room temperature} = 22^\circ\text{C} \quad (\text{the limit as } t \rightarrow \infty). \quad \text{A1}$$

$$\text{(c)} \quad 54 = 22 + 64e^{-4k} \Rightarrow 64e^{-4k} = 32. \quad \text{M1}$$

$$e^{-4k} = \frac{1}{2}. \quad \text{A1}$$

$$-4k = \ln \frac{1}{2} = -\ln 2 \Rightarrow k = \frac{1}{4} \ln 2. \quad \text{A1 AG}$$

$$\text{(d)} \quad T(10) = 22 + 64e^{-10k} = 22 + 64e^{-(5/2)\ln 2}. \quad \text{M1}$$

$$= 22 + 64 \cdot 2^{-5/2} = 22 + \frac{64}{4\sqrt{2}} = 22 + 8\sqrt{2}. \quad \text{A1}$$

$$= 33.314\dots \approx 33.3^\circ\text{C} \quad (3 \text{ s.f.}). \quad \text{A1}$$

$$\text{(e)} \quad \text{Set } T(t) = 30: 30 = 22 + 64e^{-kt} \Rightarrow 64e^{-kt} = 8 \Rightarrow e^{-kt} = \frac{1}{8}. \quad \text{M1}$$

$$-kt = -3 \ln 2 \Rightarrow t = \frac{3 \ln 2}{k} = \frac{3 \ln 2}{(1/4) \ln 2} = 12. \quad \text{M1}$$

$$t = 12 \text{ minutes}. \quad \text{A1}$$